Lumbini City College, Drivertole, Rupandehi **FM :60**

BCA 3rd semester Final Internal Assessment /Statistics and Probability / 2082 Time: 3 hrs

GROUP B: Short questions (Attempt all the questions) (6 x5 =30)

1. Write down the process of collecting Primary data
2. Estimate the marks in JAVA when the marks in statistics in 65 by using following data.

Marks in statistics	57	58	59	59	60	61	62	64
Expenses	77	78	75	78	82	82	79	81

3. Fit Bionomial Distribution from the following data where p = 0.5

No. of health	0	1	2	3	4
Frequency	28	62	46	20	4

4. What is sampling? The Standard Deviation of marks in an entrance exam in BCA students is 0.5. How large a sample must be taken in oder to be 95% confidence that the error of his/her estimate will not exceed 0.01.
5. Calculate the Coefficient of variation from from the following data:

Age(year)	10-20	20-30	30-40	40-50	50-60
No. of person	8	20	15	17	25

6. The marks distribution of 100 students of a college is given below:

Marks	10-20	20-30	30-40	40-50	50-60
No of students	15	20	30	20	15

Find the P23 and Q3 of the following data.

GROUP C (Attempt all the questions) (10 x2 =20)

7. A test was given to three candidates taken at random from three province of Nepal. the scores of candidates are given below:

Gandaki	9	7	6
Lumbini	7	4	5
Bagmati	6	5	6

Carry out one-way ANOVA

8. Student's age in the regular daytime BCCA program and the morning time BCA program of campus are described by two samples. If the homogeneity in age of the class is positive factor in learning make suggestions, with reason which of two groups will be easier to teach

Popular BCA Program		Morning BCA Program	
Age	Number of student	Age	Number of student
23	9	27	19
29	2	31	8
28	5	30	5
22	10	29	4
30	1	28	6
21	4	33	5
25	11	34	5
26	6	35	11
27	3	36	2
24	9	32	4
Total	60	Total	69

Lumbini City College

Pre-Board Examination - 2025

BCA-3rd Semester: OOP In java

Full Marks: 60 Time: 3 Hrs.

Group B

Attempt any six questions. [6*5= 30]

2. What are different types of inheritance? How multiple inheritance is achieved in java?

3. Write a program in java to create an object and store the object itself in file assuming all the member fields are non transient. Later reconstruct the object from the file.

4. What is synchronization? How do you create a synchronous block and method in java?

5. What is Applet? Draw and explain the life cycle of applet in brief.

6. What is AWT? How do you differentiate between AWT and Swing.

7. String is Immutable class. Justify it with proper reasoning. What are mutable associates of String class?

8. Differentiate between Map, List and Set with suitable example

Group c

Attempt any 2 questions. [2*10= 20]

9. Write a program to implement MDI form that creates student registration form with validation. Form should contain at least 10 fields of 5 different types. (Validation: Name is required, age must be 18+, gender must be selected, mobile number must be of 10 digits)

10. Write a menu driven program to perform below operations for the input given by user.

a. Convert Binary number to decimal
b. Decimal number to Binary

11.

a. What do you mean by constructor? Explain different constructor's types in detail.
b. What is anonymous class? Create an anonymous inner class using A: interface. B: abstract class and instantiate them separately.

Lumbini City College

Affiliated to Tribhuvan University

Subject: Web Technology
Semester: III

Full Marks: 60
Pass Marks: 24
Time: 3hrs

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.

Group B

Attempt any SIX questions 6x5=30

2) What is an image map? Describe the key steps involved in creating a client image map in HTML using necessary tags and attributes.

3) Write HTML tag to generate the following table.

Weather Station

Station	Temperature		Humidity	Weather
	Max	Min		
USA	24	19	60%	cloudy
Germany	5	-1	70%	Rainy

4) Enlist and explain method of using CSS in web page.

5) Explain the CSS Box Model and its different elements.

6) Write a form to collect details of a user such as name, address, radio button to choose subject of book he wants to buy, dropdown to choose favorite author.

7) How XSL is different from CSS? Explain.

8) Create a server side script to use session variables for the login and logout process without using database.

Group C

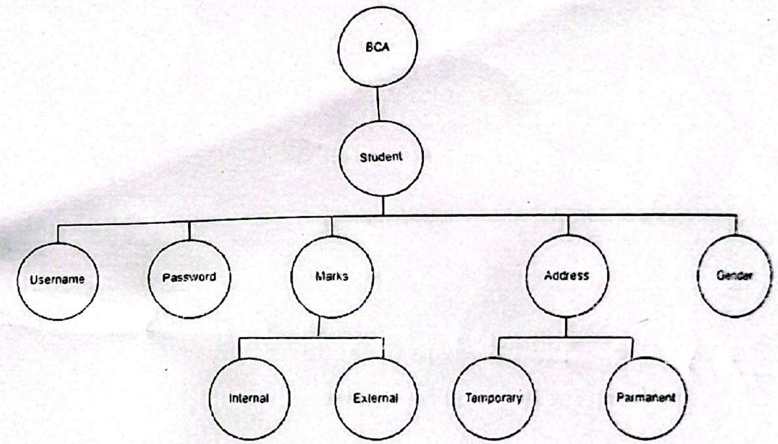
Attempt any TWO questions 2x10=20

9) Write an HTML script with CSS for following elements with effects

a) Link: after visited red color and before visited blue color.
b) Table with odd rows pink and even rows blue color.
c) Text color-green, Text-indentation-2cm, Text case-uppercase.

10) Develop a simple web page that asks the users input (name, email, phone, gender) and store into database using server side script.

11) Using the following structure, produce a well-formed XML file and use the following validation criteria to produce an XSD schema.



Rules for XSD validation:

a) The username must have a minimum of 10 characters.
b) The password must be 8 characters long and begin with Aa-Zz.
c) Internal marks should range from 0 to 60, while external marks should range from 0 to 20.
d) The gender must be one of male, female, or other.
e) The salary should range from Rs 25000 to Rs 50000.

Best of Luck

Lumbini City College

Pre-Board Examination

2025

BCA III | FM:60
System Analysis and Design | PM:24
CACS203 | Time:3hrs

Candidates are required to answer the questions in their own words as far as possible.

Group B [6*5 = 30]

Attempt any SIX questions.

2. What is system? What are the phases of SDLC? Explain briefly.

3. Why is project management important? Briefly expiain the activities performed by the project manager during project execution.

4. List various methods of interacting with a system. Briefly explain the factors to be considered while designing a form?

5. Define software testing? Briefly explain the different approaches to installation.

6. Why is normalization required? State second normal form and explain it with a proper example.

7. What are the traditional methods for determining requirements? Explain in brief.

8. What is System Maintenance? What are the different types of maintenance? Explain briefly

Group C [2*10 = 20]

Attempt any TWO questions.

9. Draw possible context and first level DFD for the following System.

A hostel has 500 rooms and 4 messes. Currently, there are 1000 students in all in 2 seated rooms. They eat in any of the messes but can get rebate if they inform and do not eat for at least 4 consecutive days. Besides normal menu, extra items are also given to students when they ask for it. Such extras are entered in an extra book. At the end for the month a bill is prepared based on the normal daily rate and extras given to each student. Systems for stores issue and control is maintained for daily use of perishables and non perishables items and order to vendor and supplies are also maintained as well.

10. What are the major difference between Agile methodologies and waterfall model? Explain about the different stages of Agile Modelling. What are its advantages and disadvantages?

11. What do you mean by feasibility analysis? Discuss about different categories of feasibility.

LUMBINI CITY COLLEGE
Drivertole, Rupandehi

Subject : Data Structures and Algorithms **Full marks 60**

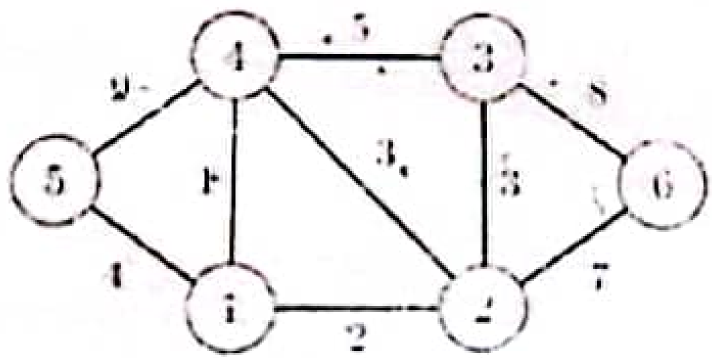
Semester : III , Paper : CACS 201 **Pass marks 24**

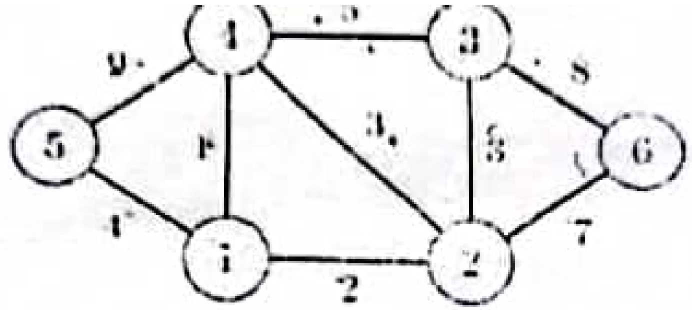
Attempt any two questions [2*10 = 20]

1. What are collision resolution techniques? Explain at least 3 collision resolution techniques.
2. Write quick sort algorithm. Sort following data using quick sort algorithm in ascending order

   12, 32, 1, 2, 5, 100, 4, 40

3. What is spanning tree? Write kruskal's algorithms for minimum spanning tree. Find minimum spanning tree of following graph.





Attempt any six questions [5*6=30]

1. Trace heap sort for data 50, 30, 70, 20, 40, 60, 80 and 23.
2. Explain Huffman algorithms showing its efficiency for text "lee bea third semester".
3. Write binary search algorithm. Trace binary search algorithms to search key element 23 in array 2 ,3 ,5, 7, 8, 25,26, 29.
4. What is hashing? Explain different hashing method.
5. Write an algorithm for infix to prefix conversion and evaluation.
6. What is asymptotic notations? Explain with example.
7. Construct a Binary Search Tree (BST) for the given sequence of numbers: {50, 30, 70, 20, 40, 60, 80}. Show step-by-step insertion and draw the final BST.